

Exercise 2 Stored Procedures

-- Create Database

```
CREATE DATABASE bank_procedures;  
USE bank_procedures;
```

-- Tables

```
CREATE TABLE savings_accounts (  
    account_id INT PRIMARY KEY,  
    customer_id INT,  
    balance DECIMAL(12,2)  
);
```

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(100),  
    department_id INT,  
    salary DECIMAL(12,2)  
);
```

```
CREATE TABLE accounts (  
    account_id INT PRIMARY KEY,  
    customer_id INT,  
    balance DECIMAL(12,2)  
);
```

-- Sample Data

```
INSERT INTO savings_accounts VALUES  
(501,1,20000),  
(502,2,15500),  
(503,3,9800);
```

```
INSERT INTO employees VALUES  
(201,'Aniket More',10,45000),  
(202,'Kavita Rane',10,52000),  
(203,'Sameer Pawar',20,48000);
```

```
INSERT INTO accounts VALUES  
(701,1,10000),  
(702,2,3000);
```

```
-- =====  
-- Procedure 1 : Monthly Interest  
-- =====
```

```
DELIMITER //
```

```
CREATE PROCEDURE ProcessMonthlyInterest()  
BEGIN  
    UPDATE savings_accounts  
    SET balance = balance + (balance * 0.01);  
END //
```

```
DELIMITER ;
```

```
-- =====  
-- Procedure 2 : Employee Bonus  
-- =====
```

```
DELIMITER //
```

```
CREATE PROCEDURE UpdateEmployeeBonus(  
    IN p_department_id INT,  
    IN p_bonus_percent DECIMAL(5,2)  
)  
BEGIN  
    UPDATE employees  
    SET salary = salary + (salary * p_bonus_percent / 100)  
    WHERE department_id = p_department_id;  
END //
```

```
DELIMITER ;
```

```
-- =====  
-- Procedure 3 : Transfer Funds  
-- =====
```

```
DELIMITER //
```

```
CREATE PROCEDURE TransferFunds(  
    IN p_from_account INT,  
    IN p_to_account INT,  
    IN p_amount DECIMAL(12,2)  
)
```

```

BEGIN
    DECLARE v_balance DECIMAL(12,2);

    SELECT balance
    INTO v_balance
    FROM accounts
    WHERE account_id = p_from_account;

    IF v_balance >= p_amount THEN

        UPDATE accounts
        SET balance = balance - p_amount
        WHERE account_id = p_from_account;

        UPDATE accounts
        SET balance = balance + p_amount
        WHERE account_id = p_to_account;

        SELECT 'Transfer Successful' AS Message;

    ELSE

        SELECT 'Insufficient Balance' AS Message;

    END IF;
END //

DELIMITER ;

-- =====
-- Execute Procedures
-- =====

CALL ProcessMonthlyInterest();

CALL UpdateEmployeeBonus(10,5);

CALL TransferFunds(701,702,2000);

-- =====
-- Results
-- =====

SELECT * FROM savings_accounts;

```

```
SELECT * FROM employees;
```

```
SELECT * FROM accounts;
```